

AI FOR DATA VISUALISATION

Q1. What is AI in Data Visualization? Explain how it differs from traditional data analysis.

AI in data visualisation refers to the use of machine learning, natural language processing and automation to analyse data and present insights through interactive charts, graphs, dashboard and visual reports .

Difference from Traditional Data Analysis:

- AI analysis is automated and predictive.
- Traditional analysis is mostly manual and based on past data.

Q2. Explain how AI helps in automatic insight generation in BI tools.

AI helps BI tools by:

- Detecting patterns and trends
- Predicting future outcomes
- Identifying anomalies
- Generating automatic reports and dashboards
- Supporting natural language queries

Q3. What are the key AI features available in Microsoft Power BI? Explain any two with examples.

In Microsoft Power BI, there are several AI features such as Q&A, Key Influencers, Smart Narratives, Forecasting, and Anomaly Detection. One important feature is Q&A, where users can type questions in natural language like 'Show total sales by month,' and Power BI automatically creates the required visualization. Another feature is Key Influencers, which helps identify the main factors affecting a result. For example, it can show that discounts or advertising campaigns are increasing sales.

Q4. Explain the concept of Key Influencers and how AI identifies factors affecting a metric.

Key Influencers is an AI feature in Microsoft Power BI that helps identify the main factors affecting a business metric or outcome.

AI analyzes the dataset using machine learning algorithms and finds patterns, relationships, and variables that have the biggest impact on the selected metric.

For example, if sales are increasing, Power BI may identify factors like higher discounts, online marketing, or customer region as the key influencers.

This feature helps businesses understand the reasons behind trends and make better decisions.

Q5. What is the role of Smart Narratives in AI-powered dashboards?

Smart Narratives in Microsoft Power BI automatically generate text summaries from dashboard data using AI. Its role is to explain key insights, trends, and performance in simple language without manual reporting. For example, Smart Narratives can automatically display statements like 'Sales increased by 15% this month compared to last month.'

Q6. Explain how Tableau Pulse provides proactive insights in Tableau.

Tableau Pulse is an AI-powered feature that provides proactive and personalized insights automatically. It continuously monitors business metrics and alerts users about important changes, trends, or anomalies without needing manual analysis. For example, Tableau Pulse can notify users if sales suddenly drop or if customer engagement increases significantly.

Q7. What is meant by AI-driven decision-making? How does AI support business decisions?

AI-driven decision-making means using Artificial Intelligence to analyze data, generate insights, and help businesses make smarter and faster decisions.

AI supports business decisions by:

- Identifying patterns and trends
- Predicting future outcomes
- Detecting risks or anomalies
- Providing real-time insights
- Automating data analysis

For example, AI can predict customer demand or identify which products generate the highest sales, helping companies improve planning and strategy

Q8. Explain the importance of interpreting AI-generated insights instead of directly trusting them.

AI-generated insights should be interpreted carefully instead of directly trusting them because AI predictions depend on the quality and accuracy of data.

Sometimes AI may give incorrect, biased, or incomplete results if the data has errors or missing values.

Human analysis is important to:

- Verify the insights
- Understand business context
- Avoid wrong decisions
- Ensure accuracy and reliability

For example, AI may predict low sales due to incomplete data, but a manager can identify external factors affecting the result

Q9. Discuss the limitations of AI in data visualization tools.

AI in data visualization tools has some limitations.

First, AI depends heavily on data quality. Incorrect or incomplete data can produce wrong insights.

Second, AI may sometimes generate biased or inaccurate predictions.

Third, complex AI models can be difficult to understand and interpret.

AI tools also require skilled users, proper training data, and high computing resources.

Finally, AI cannot fully replace human judgment because business decisions still need human understanding and experience.

Q10. Compare how AI is used differently in Power BI and Tableau .

Microsoft Power BI and Tableau both use AI to improve data analysis, but their focus is slightly different.

Power BI focuses more on AI features like Q&A, Key Influencers, Smart Narratives, and integration with Microsoft AI services for automated analysis.

Tableau focuses more on visual analytics and proactive insights through features like Tableau Pulse, Explain Data, and smart visual exploration.

In simple terms, Power BI is stronger in AI-driven business analysis and automation, while Tableau is stronger in interactive visualization and insight discovery.